MIL-R-11337F

15 May 1970
SUPERSEDING
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MILITARY SPECIFICATION

RANGES, PACKAGING AND PACKING OF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers optional detailed requirements or performance requirements for preservation, packaging, and packing for shipment and storage of household, field, and heavy duty hotel-type cooking ranges, and floor-type deep fat fryers, griddles, and baking and roasting ovens, heated by electricity, gas, gasoline, oil, wood, or coal.

1.2 Classification.-

1.2.1 Options (see 3.3).-

Option I - Detailed requirements.

Option II - Performance requirements.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip (Polyolefin).
NN-P-71	-	Pallets, Material Handling, Wood, Double Faced, Stringer Construction.
QQ-S-781	-	Steel, Strapping, Flat.

FSC 7310

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UU-P-134	-	Paper, Wrapping, Wet-Waxed.
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
UU-T-81	-	Tags, Shipping and Stock.
MMM-A-260	-	Adhesive, Water-Resistant (For Sealing Waterproofed Paper).
PPP-B-591	-	Boxes, Fiberboard, Wood Cleated.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-1163	-	Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated.
PPP-B-1364	-	Box, Corrugated Fiberboard, High Strength, Weather-Resistant, Double-Wall.
PPP-C-650	-	Crates, Wood, Open and Covered.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requriements for Packaging and Packing.
PPP-S-760	-	Strapping Nonmetallic, (and Connectors).
PPP-T-45	-	Tape, Gummed Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-97	-	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-C-104	-	Crates, Wood; Lumber and plywood Sheathed, Nailed and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-3774	-	Crates, Wood; Open, 12,000 and 16,000 Pound Capacity.
MIL-C-11133	-	Crate, Wood, Open, Wirebound.
MIL-C-46842	-	Cushioning material, Unicellular Polyethylene Foam (for Packaging Purposes).

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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DRAWING

ARMY NATICK LABORATORIES

5-11-795 - Ranges; Containers for level A and level B Shipment.

(Miniature copy of Drawing No. 5-11-795, identified as figure 1, is attached for information purposes only.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Material shall be as specified herein.

3.2 Disassembly and matchmarking.-

3.2.1 Disassembly.- Detachable leveling pads, drip trays, racks, burner grates, broiler pans and accessories shall be removed. Surface heating units or burners and oven and broiler heating units shall either be removed or cushioned with material specified in 3.4.1.1.7, or with corrugated fiberboard, and secured in their respective positions with either clips or with wire of not less than 16 gage, or blocked or braced in such a manner as to prevent damage

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during shipment. Projecting parts, easily removed, may be detached from the unit for the reduction of cubage. Nuts, bolts, and screws removed during disassembly shall be reinserted and secured in their respective locations. Alternatively, the nuts, bolts, and screws may be placed in a cloth drawstring bag, the bag cushioned and securely attached in a conspicuous place on the unit in a manner so as not to damage the finish of the unit.

3.2.2 Matchmarking.- Cloth shipping tags conforming to type A, size 1 of UU-T-81 shall be attached securely to the parts removed and to mating parts on the unit in such a manner as not to damage the item or the preservation thereof. Suitable matchmarking and identification shall be printed on these tags.

* 3.3 Selection of options.- Units, having a combined gross weight (complete unit and shipping container) not exceeding 800 pounds, shall be preserved, packaged and packed either option I or II, at the discretion of the contractor. Units with a combined gross weight exceeding 800 pounds shall be preserved and packaged in accordance with option I.

3.4 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

* 3.4.1 Level A.-

3.4.1.1 Option I, detailed requirements (see 3.3).- The cleaning process, the application and types of preservatives shall be in accordance with MIL-P-116 as specified herein.

3.4.1.1.1 Cleaning.- All exposed, uncoated, unpainted ferrous metal surfaces of the unit, accessories and parts shall be thoroughly cleaned by process C-1.

* 3.4.1.1.2 Repainting and recoating.- Painted and permanently coated surfaces from which the paint and permanent coating have been removed, or which are defective, shall be cleaned, treated, and repainted or recoated in accordance with the original requirements.

3.4.1.1.3 Application of preservative.- All cleaned, uncoated, and unpainted ferrous metal surfaces, except corrosion-resisting steel, shall be coated with type P-14 preservative.

* 3.4.1.1.4 Protection of preservative.- All exposed surfaces of the units, accessories, and parts, coated with preservatives that might contact other surfaces, including packaging materials, shall be covered, wrapped or bagged with greaseproof barrier material conforming to type II, grade A, class 1 or 2 of MIL-B-121. Covering, wrapping or bagging shall be heat sealed or secured with tape conforming to PPP-T-45 or jute, sisal, manila or cotton twine.

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3.4.1.1.5 Operating instructions.- Operating instructions and service manuals shall be individually packaged in accordance with method IC-1 or method IC-3 and securely attached to the unit.

3.4.1.1.6 Accessories and parts.- Component parts and accessories, disassembled as specified in 3.2, shall be cushioned by wrapping in cushioning materials, as specified in 3.4.1.1.7 and then packaged in a fiberboard box or boxes conforming to type CF, class domestic, variety SW, grade 200, style RSC of PPP-B-636. Each fiberboard box shall be securely closed in accordance with the appendix of the box specification. The fiberboard box or boxes shall be secured either within the unit or in an unused section of the shipping container.

* 3.4.1.1.7 Doors and drawers.- All doors and drawers of the unit shall be cushioned in a closed position with cushioning material conforming to type II, class B of PPP-C-843 or MIL-C-46842. The cushioning material shall be placed between the body of the unit and points of contact of the doors, drawers, and other panels to prevent chipping of the porcelain enamel finish. The doors and drawers shall be secured in the closed position with steel strapping 1/4 by 0.015 inch on the household units and 3/8 by 0.020 inch on the heavy duty, or hotel-type unit. Strapping shall conform to type I, class A, of QQ-S-781. A continuous strip of cushioning material conforming to type II, class B of PPP-C-843 shall be placed under the steel strapping to prevent contact of the steel straps with the porcelain or enameled surfaces of the unit. The strip of cushioning material shall be minimum 1/4 inch thick and 1-1/2 inches wide, completely engirdling the unit and shall be centered under the steel strapping. Alternatively, the doors and drawers may be secured in closed position with minimum 1/2 inch width adhesive tape on all household type units and minimum 1 inch width tape on all heavy duty or hotel-type units and floor-type fryers, baking and roasting ovens. The tape shall conform to type III of PPP-T-97 and shall show not staining when tested in accordance with 4.4.1. The tape shall be applied firmly with a hard rubber roller and shall extend not less than 6 inches on the door or drawer and onto each adjacent end or side surface no less than 6 inches. In lieu of steel strapping and tape banding, nonmetallic strapping conforming to type II or III of PPP-S-760 may be used provided that the breaking strength is 310 pounds for household units, and 420 pounds for heavy duty or hotel type. When nonmetallic strapping is used no cushioning is required and seals shall be applied on the back surface of the unit.

3.4.1.2 Option II, performance requirements (see 3.3).-

3.4.1.2.1 Preservation.- Preservation shall be as specified in 3.4.1.1 through 3.4.1.1.4.

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* 3.4.1.2.2 Packaging.- Accessories, parts, doors, and drawers shall be cushioned and packaged in a manner to meet the performance test requirements for the applicable packing level of PPP-P-600.

* 3.4.2 Level C.- Each unit shall be cleaned, preserved and packaged to afford adequate protection against deterioration and damage during shipments from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

3.5 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 3.5.1 Level A.-

3.5.1.1 Option I, detailed requriements (see 3.3).- Each unit, together with accessories and parts, preserved and packaged as specified in 3.4.1.1 or 3.4.2 shall be packed as indicated in table I (see 3.5.4) for weight limitations and applicable securing methods, in shipping containers specified in 3.5.1.1.1, 3.5.1.1.2 or 3.5.1.1.3, as applicable.

* 3.5.1.1.1 Box, fiberboard, maximum weight of contents 65 pounds.- The fiberboard shipping containers shall be snug fitting conforming to style optional V2s or V3s of PPP-B-636. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of the type CF, variety SW, class weather-resistant, grade W5c, A or C flute of PPP-F-320. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with bands in accordance with the appendix of the container specification. The maximum weight of contents of V3s shall be 40 pounds and V2s, 65 pounds.

* 3.5.1.1.2 Boxes, wood cleated plywood, maximum weight of contents 1000 pounds.- Wood cleated plywood shipping containers shall conform to overseas type of PPP-B-601, and as shown in Drawing No. 5-11-795.

* 3.5.1.1.3 Crates, wood; lumber and plywood sheathed, nailed and bolted, weight of contents exceeding 1000 pounds.- Sheathed nailed wood crates shall conform to type I or II, class 1 or 2, style a of MIL-C-104.

* 3.5.1.2 Option II, performance requirements (see 3.3).- Each unit preserved and packaged as specified in 3.4.1.2 or 3.4.2 shall be packed in accordance with the level A requirements of PPP-P-600. The packed unit shall meet the level A performance requirements of PPP-P-600.

* 3.5.2 Level B.-

* 3.5.2.1 Option I, detailed requirements (see 3.3).- Each unit, together with accessories and parts, preserved and packaged as specified in 3.4.1.1 or 3.4.2 shall be packed as indicated in table I (see 3.5.4) for weight limitations and applicable securing methods, in shipping containers specified in 3.5.2.1.1 through 3.5.2.1.7, as applicable. The contents of each open type

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crate shall be protected from the elements by a full length shroud fabricated of minimum 60 pound kraft wrapping paper conforming to grade A of UU-P-268. Seams shall be sealed with water-resistant adhesive conforming to MMM-A-260. Alternatively, the shroud shall be fabricated from nominal 3 mil polyethelene conforming to L-P-378. All seams of the polyethelene shroud shall be heat sealed. The shroud shall hang free, when possible, and shall extend to the floor of the crate base. All sharp points of contact between the item, the blocking and bracing, and the shroud shall be cushioned to prevent rupture or chafing of the shroud. All open type crates shall be assembled, closed and strapped in accordance with the appendix of the applicable crate specification.

* 3.5.2.1.1 Box, fiberboard, maximum weight of contents 120 pounds.- The fiberboard shipping containers shall be snug fitting conforming to type CF or SF, class domestic, style optional of PPP-B-636. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of type CF, variety SW or DW, class domestic, minimum grade 200 of PPP-F-320. Each fiberboard shipping container shall be securely closed in accordance with the appendix of the container specification. When specified (see 6.2) the fiberboard shipping containers shall be fabricated of material conforming to V3c, V3s or V4s of PPP-F-320. The weight of contents shall not exceed 65 pounds for V3c, V3s or V4s boxes and the normal weight requirements of PPP-B-636 for class domestic boxes.

* 3.5.2.1.2 Boxes, fiberboard, corrugated, triple-wall, maximum weight of contents 275 pounds.- Triple-wall corrugated fiberboard boxes shall conform to class 1, grade A, style E, of PPP-B-640. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of material specified in 3.5.2.1.1, except that the minimum grade shall be 275. Each fiberboard shipping container shall be securely closed in accordance with the appendix of the container specification. When specified (see 6.2), the triple-wall corrugated shipping containers shall be fabricated of class 2, weather-resistant material. When weights of contents and containers exceed 250 pounds the container shall be strapped to a pallet conforming to NN-P-71 of size to fit the base of the container. Strapping used shall be in accordance with the appendix of PPP-B-640. Alternatively, double wall fiberboard boxes conforming to type DW, grade 600 of PPP-B-1163 or PPP-B-1364 may be used.

3.5.2.1.3 Boxes, fiberboard, wood cleated, maximum weight of contents 400 pounds.- Wood cleated fiberboard shipping containers shall conform to class 1 of PPP-B-591, and as shown in Drawing No. 5-11-795.

* 3.5.2.1.4 Boxes, wood, cleated-plywood, maximum weight of contents 1000 pounds.- Wood cleated plywood shipping containers shall conform to domestic type of PPP-B-601 and as shown on Drawing No. 5-11-795.

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3.5.2.1.5 Crate, wood, open, wirebound, maximum weight of contents 1000 pounds.- Wire bound crates shall conform to type II, class 1 or 2, style 1 or 2, grade B; or type III, class 1, 2 or 3, style 1 or 2, grade B of MIL-C-11133, as applicable.

3.5.2.1.6 Crates, wood, open and covered, maximum weight of contents 2,500 pounds.- Open wood crates shall conform to style A, type I, II, IV, or V, class 1 of PPP-C-650, as applicable.

3.5.2.1.7 Crates, wood; open, 12,000 and 16,000 pound capacity, contents exceeding 2,500 pounds.- Open wood crates shall conform to type I, style A, of MIL-C-3774.

* 3.5.2.2 Option II, performance requirements (see 3.3).- Each unit, preserved and packaged as specified in 3.4.1.2 or 3.4.2, shall be packed in accordance with the level B requirements of PPP-P-600. The packed unit shall meet the level B performance requirements of PPP-P-600.

* 3.5.3 Level C.- Units, together with accessories and parts, preserved and packaged as specified in 3.4, shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

3.5.4 Securing of units in the shipping container.- Each unit shall be secured within the shipping container by the methods as indicated in table I for the weight limitations specified. A minimum of 1 inch clearance shall be provided between the top, sides and ends of the unit and the adjacent inside surfaces of the shipping container.

TABLE I.- Methods of securing units in shipping containers and maximum weight of contents

Container	Securing method	Weight of contents (pounds)
Box, Fiberboard, PPP-B-636	See 3.5.1.1.1 and 3.5.2.1.1	Unless otherwise specified the weight of contents shall not exceed the normal weight limitation of the specification.
Boxes, Fiberboard, Corrugated, PPP-B-640, PPP-B-1163 or PPP-B-1364	See 3.5.2.1.2	275 maximum

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TABLE I.- Methods of securing units in shipping containers and maximum weight of contents (cont'd)

Container	Securing method	Weight of contents (pounds)
Boxes, Fiberboard, Wood Cleated a/ PPP-B-591 and Drawing No. 5-11-795	2	300 maximum
	1 or 3 or 1 and 3	400 maximum
Boxes, Wood, Cleated Plywood, PPP-B-601 and Drawing No. 5-11-795	2 b/	300 maximum
	1 or 3 b/ or 1 and 3 b/	1000 maximum
Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted, MIL-C-104	1 or 3 b/ or 1 and 3 b/	Exceeding 1000
Crate, Wood, Open, Wirebound, MIL-C-11133	2	300 maximum
	1 or 3 or 1 and 3	1000 maximum
Crates, Wood, Open and Covered, PPP-C-650	2	300 maximum
	1 or 3 or 1 and 3	2,500 maximum
Crates, Wood; Open 12,000 and 16,000 Pound Capacity, MIL-C-3774	1 or 3 or 1 and 3	Exceeding 2,500

a/ For level B shipments only.

b/ Cushioning material shall be water-resistant for level A shipment.

3.5.4.1 Method 1 (bolting).- Units, fabricated with provision for bolting, shall be secured to the base of the container as indicated in 3.5.4.1.1.

* 3.5.4.1.1 Base.- The unit shall be shipped upright and secured to the base of the shipping container with carriage or machine bolts extending through the load bearing members of the base or through the skids and the load bearing members of the base into the unit base or legs. The heads of the bolts shall be at the underside of the load bearing members or the skids, and not less than four bolts shall be used for securing the unit to the crate base. Carriage bolt diameters shall be minimum five-sixteenth of an inch for units weighing less than 400 pounds, and three-eights of an inch for units weighing not more than 1,500 pounds. Units in excess of 1,500 pounds shall require 1/2 inch diameter carriage bolts. Round flat washers shall be used under the heads of all machine bolts, and lockwashers shall be used under all nuts.

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* 3.5.4.2 Method 2, pressure pack.- The unit shall be cushioned and held securely within the shipping container by built-up single wall, double wall or triple wall corrugated fiberboard edge pads or cushioning material conforming to MIL-C-46842 of adequate thickness to provide a minimum of 1 inch clearance between any protruding part of the unit and the interior surface of the shipping container, and shall be of adequate width to provide a minimum width contact area of 3 inches on all horizontal edges at top, bottom and sides, and vertical end edges of the unit. When the splash back is an integral part of the unit, a sheet of double-walled corrugated fiberboard or cushioning material conforming to MIL-C-46842 having a minimum of 1/4 inch thickness shall completely cover the top of the unit and shall have a top locking (hold-down) consisting of half-slotted partitions, spacing not exceeding 4 inches, fabricated of 275 pound test double-walled corrugated fiberboard placed on top of the pad covering the top of the unit. The partitions shall be of adequate height to provide a minimum of 1 inch clearance above the top of the splash back. In lieu of the pad and half-slotted top blocking (hold-down), any similar corrugated fiberboard blocking assembly or cushioning may be used provided it protects the top of the unit and allows a minimum of 1 inch clearance above the splash back. Method B shall not be used for units weighing more than 300 pounds. At points of contact with porcelain enamel surfaces, corrugated fiberboard shall be provided with nonabrasive wax coating (see 3.5.5.c).

* 3.5.4.3 Method 3, blocking and bracing.- The unit shall be secured within the container by means of wood blocking and bracing. The blocking and bracing shall be cushioned at all points of contact with the unit with approximately 1/4 inch thick cellulose wadding or foam conforming to type II, class B of PPP-C-843 or MIL-C-46842, respectively. Blocking and bracing shall not have any bearing on any surface of the unit, subject to damage by distortion or abrasion. The blocking and bracing shall not depend on end grain nailing to the container, but shall be supplemented with supporting blocks and cleats.

* 3.5.5 Protection of enamel or porcelain surfaces for method B or C.- One of the following shall be placed between the cushioned blocking or corrugated fiberboard pads and the enamel or porcelain surfaces of the unit:

(a) A sheet of wax paper conforming to type III of UU-P-134.
(b) Cellulosic cushioning material having a thickness of not less than 0.120 inch conforming to type II, class B of PPP-C-843.
(c) Corrugated fiberboard with nonabrasive wax coating.
(d) A sheet of nominal 3 mil polyethelene film conforming to L-P-378.

3.6 Marking

3.6.1 Interior packages.- Interior packages shall be marked in accordance with MIL-STD-129. Component parts that have been overwrapped and secured in their respective locations will not require marking.

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3.6.2 Shipping containers.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

3.6.2.1 Precautionary marking.- Special handling marking requirements in MIL-STD-129 applicable to fragile items, such as arrows, the words "UP" and "THIS SIDE UP", and "CENTER OF BALANCE", shall apply.

3.7 Workmanship.- Preservation, packaging, packing and marking shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Components and materials inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification.

4.2.1.1 Tape shall be tested in accordance with 4.4.1 through 4.4.1.3, to determine compliance with 3.4.1.1.7. Any evidence of staining will be cause for rejection of the tape.

* 4.3 Examination of the end item.- Examination of the end item, preparations of packaging, packing, and marking shall be in accordance with the classification of defects set forth in tables II, III, IV, V, and VI, and the inspection levels and acceptable quality levels (AQLs) specified in 4.3.1. The sample unit for this examination shall be one disassembled packaged or packed range, as applicable, of the specified type, design and characteristics. The lot size shall be expressed in terms of the applicable sample unit for purpose of determining sample size.

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* TABLE II.- Visual examination for preparation for packaging

Examine	Defect	Category Major	Minor
Disassembly	Detachable parts and accessories not removed (when specified)	101	
	Detachable parts and accessories not cushioned and secured in position or bagged and cushioned (when specified)	102	
	When nuts, bolts and screws are bagged the bag is not cushioned, or secured in a conspicuous place	103	
Matchmarking	Tags with information not attached to mating parts		201
Repainting and recoating	Not refinished or recoated as needed	104	
	Material not identical to original	105	
Preservative coating	Barrier missing or not covering coating at contact points	106	
	Barrier not taped or tied in position	107	

* TABLE III.- Examination of the end item for packaging

Examine	Defect	Category Major	Minor
Option I - Levels A and C			
Operating instructions and service manual	Not placed individually in envelope		202
	Not securely attached to unit		203
	Wrong method of packaging	108	
Door and drawers	Not secured as specified	109	
	Incorrect width and thickness of cushioning	110	
	Not cushioned as specified	111	
	Cushioning material missing (when specified)	112	

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* TABLE III.- Examination of the end item for packaging (cont'd)

Examine	Defect	Category Major	Category Minor
Doors and drawers (cont'd)	Incorrect size of strap	113	
	Strap not centered on cushioning material when required		204
	Strap tensioned too tightly causing damage	114	
	Strap tensioned too loosely causing slippage	115	
	Not taped as specified (when required)	116	
Accessories and parts (boxed)	Not wrapped with cushioning material	117	
	Not packaged in box specified	118	

* TABLE IV.- Examination of the end item for packing

Examine	Defect	Category Major	Category Minor
Option I - Levels A, B and C			
Range	Not secured and cushioned within container as specified (see 3.5.1.1.1, 3.5.1.1.2, 3.5.1.1.3, and 3.5.4)	119	
	Finished surfaces not protected from abrasion with material specified	120	
Accessories and parts box	Not securely blocked and braced within range (when specified)	121	
	Not secured within the shipping container (when specified)	122	
Container	Not as specified for weight limits of range (see table I)	123	
	Less than 1 inch clearance specified between range and inside of container	124	

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* TABLE IV.- Examination of the end item for packing (cont'd)

Examine	Defect	Category Major	Minor
Shroud	Shroud missing (when specified)	125	
Bolts and nuts	Less than four bolts holding range when required	126	
	Diameter of bolt less than specified when required	127	
	One or more flat washers missing under heads of machine bolts when required	128	
	One or more nuts loose on bolts	129	
	One or more lockwashers missing	130	
	Bolts not positioned through container members and blocking as specified	131	
Blocking members	Incorrectly placed	132	
Edge pad	Not built-up of material specified	133	
	Less than length required to cover edge	134	
	Less than three inches of pad contacting each adjacent corner surface of range	135	
	Does not provide minimum one inch clearance beyond outermost projection	136	
	Not fitting as requred to make tight pack	137	
Tray	Not double faced corrugated when required	138	
	Less than 4 inches deep	139	
	Missing on top	140	

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* TABLE V.- Examination of the end item for marking

Examine	Defect	Category Major	Minor
Marking Package and shipping container (as applicable)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.	141	

4.3.1 Inspection levels and acceptable quality levels (AQLs) for examination.- The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per one hundred units shall be as follows:

Table No.	Inspection levels	AQLs Major	Total
II	S-2	2.5	10.0
III	S-2	2.5	10.0
IV	S-2	2.5	
V	S-2	2.5	
VI	S-2	2.5	4.0

4.3.2 Examination of shipping container.- When shipping containers are required to be in accordance with PPP-B-636, PPP-B-640, and MIL-C-104, examination for closure and strapping shall be in accordance with the applicable specification.

* 4.3.2.1 When shipping containers are required to be in accordance with PPP-B-601 examination of closure and strapping for determination of compliance with that specification, as applicable, and Drawing No. 5-11-795 of this specification shall be in accordance with the defects in table VI.

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* TABLE VI.- Examination for strapping and closure of the shipping container conforming to PPP-B-601

Examine	Defect	Category Major	Minor
Strapping	Not applied as specified (see Drawing No. 5-11-795)	142	
	Missing strap	143	
	Loose strap	144	
	Torn strap	145	
	Cut strap	146	
	Not drawn tight	147	
	Weak joint	148	
Staples	Staples less than 4 inches from edge of member		205
	Less than size specified	149	
	Not spaced as specified		206

4.4 Tests.-

4.4.1 Staining test.- The staining test shall be performed as specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3.

4.4.1.1 Conditioning.- Tape shall be conditioned for testing for not less than 24 hours at 73 ± 2 F. and 50 ± 4 percent relative humidity.

4.4.1.2 Specimens.- The first three layers of tape shall be removed from the rolls before taking specimens to be tested. An equal number of 10 inch lengths of tape shall be taken as specimens from all sample rolls to be tested.

4.4.1.3 Staining.- Specimens of tape shall be applied adhesive down to a clean porcelain enamel surface panel and rolled down with two passes, once in each longitudinal direction, using a 4-1/2 pound rubber roller, having a surface durometer hardness between 70 and 80, and approximately 3-1/2 inches in diameter, at a rate of travel, 10 inches per minute. Panels shall be placed in an oven having a temperature of 150 ± 2 F. for a period of 10 days. Upon removal, panels shall be allowed to cool 2 hours before stripping off the tape. Tape edge transfer may be noticed due to the high temperature. If so, it should be removed with solvent prior to determination for staining. Tape shall be rejected for use if staining is evident.

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5. PREPARATION FOR DELIVERY

5.1 Not applicable.-

6. NOTES

6.1 Intended use.- The preservation, packaging, packing and marking requirements of this specification are intended for the preparation for delivery of ranges and related items.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing (see 3.4 and 3.5).
(c) Whether class weather-resistant material shall be used for level B packing (see 3.5.2.1.1 and 3.5.2.1.2).

* 6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - SM
Navy - SA, YD
Air Force - 84

User activity:

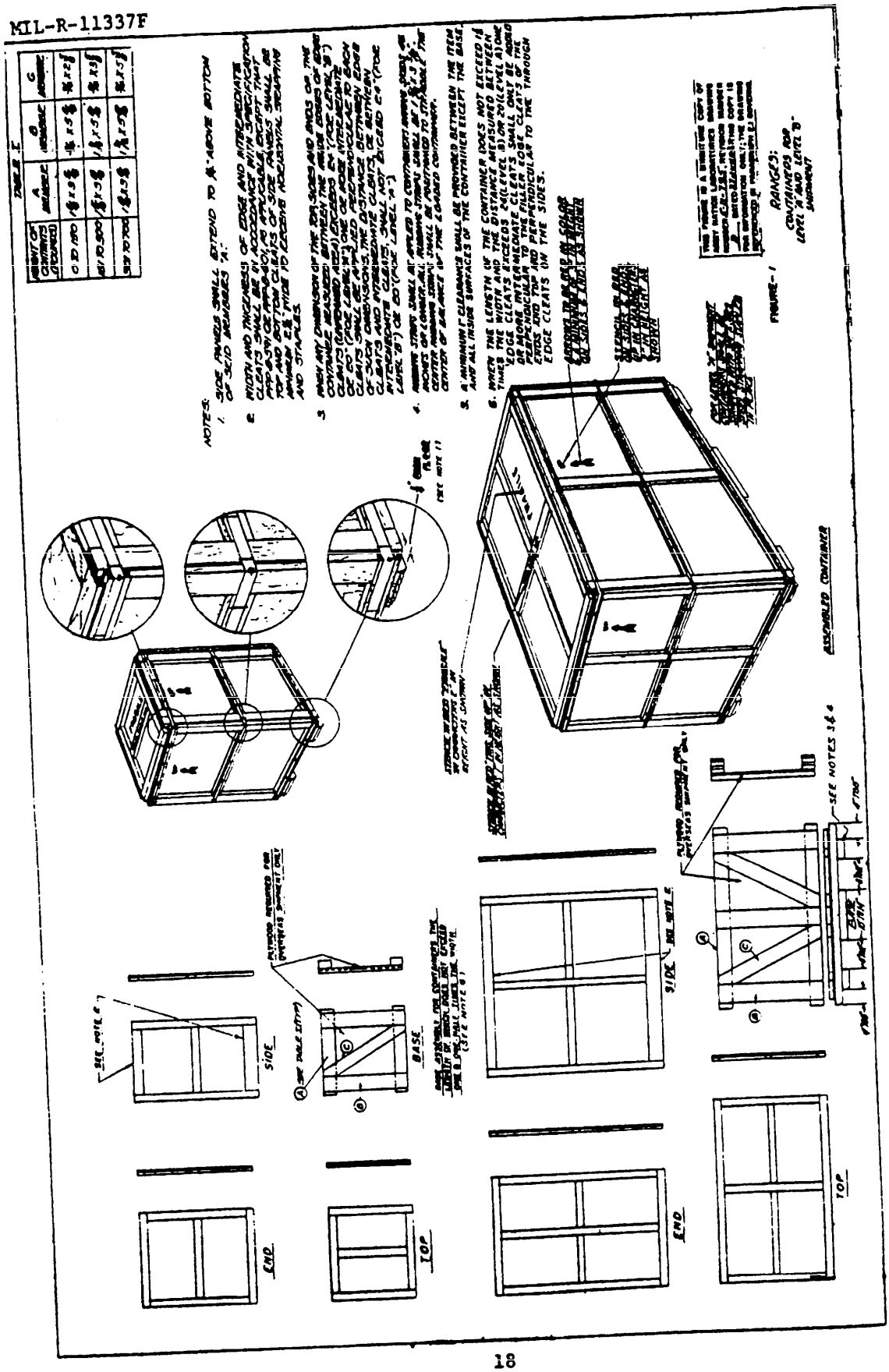
Navy - MC

Preparing activity:

Army - GL

Project No. 7310-0311

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-11337F RANGES, PACKAGING AND PACKING OF

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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